



Comparative Analysis of GenAI Models for Extensions and Validation of EV Battery Characterization Data

Sandip Sing, Rushikesh Pawar, and Umesh N. Hivarkar FEV.io

Citation: Sing, S., Pawar, R., and Hivarkar, U.N., "Comparative Analysis of GenAI Models for Extensions and Validation of EV Battery Characterization Data," SAE Technical Paper 2024-28-0032, 2024, doi:10.4271/2024-28-0032.

Received: 23 May 2024

Revised: 24 May 2024

Accepted: 29 Jul 2024

Abstract

Rapid advancement of electric vehicle (EV) technology has propelled the need for reliable and efficient methods of battery data. This has vital importance – to ensure safety aspects and efficient design of EV system. Traditional data collection methods for battery characterization is a large subject for the design of experiments and is often expert's skill intensive, time-consuming, and do not allow scalability. This study proposes an approach which bases on Generative Artificial Intelligence (GenAI) for two activities. First, to assist the DOE in characterizing cell/batteries at different C-rates and temperatures considering different degradation rates. Second, for manipulation of characterization data taking into account measurement and data recording errors. The study compares GenAI models like Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and transformer-based (Time-GPT) models in generating and validating EV battery characterization data. This is not a complete replacement for battery testing, as batteries must physically undergo cyclic aging and other tests. The paper explores ability of different

GenAI models to accurately capture critical electro-chemical and thermal features. This enables better planning of next characterization experiments and assist in eliminating boundary and intermediate scenarios for cell characterization experiments by generating synthetic data from GenAI models. The model so tuned for cell characteristics can also be extended as data manipulator to re-establish battery characterization data generated from testing experiments, considering sensor issues, data logging issues, data transportation and synchronization issues, etc. The robustness of these models in handling diverse, heterogeneous, and asynchronous datasets sourced from different EV manufacturers, battery chemistries, and specifications are scrutinized. The performance of the models is compared across multiple attributes like execution times, computing resource requirements, accuracy and consistency of generated data, and volume of data required to optimize the models. This study contributes to improve the modeling, simulation, and optimization of EV batteries and enable rapid development of data-driven products specifically for battery health analysis.

Introduction

The vast landscape of Electric Vehicle (EV) underscores the need to improve operational efficiency and precision of user-facing features. Conventional methodologies for battery characterization are robust, but often require specialized expertise and significant time investment. Moreover, their scalability may be limited, hindering the dynamic evolution of EV technology. In this context, novel technologies offer promising opportunities to synthesize the required data to complement traditional methods.

This work advocates a collaborative approach, leveraging Generative Artificial Intelligence (GenAI) to enhance battery parameter predictions and improve accuracy by generating additional data from existing datasets [1]. The study aims to extend its exploration into the domain of battery data requirements. As part of this extension, the research endeavors to delve into cell modeling and

battery health prediction, primarily focusing on physics-based modeling. The acquisition of ample data from cell testing is crucial for these endeavors; however, it typically entails a considerable time investment ranging from 10 to 14 months. Given these time constraints, it becomes imperative to initiate the implementation of physics-based models at the earliest feasible stage, even with access to only 10-15% of the test data. To expedite this process, the utilization of synthetic data generation via the GenAI engine is proposed, enabling the estimation of additional cycles in a continuous and systematic manner. Subsequently, as more data becomes available from testing, both the AI engine and physics-based models can be further refined and fine-tuned to enhance accuracy and predictive capabilities.

Numerous investigations have explored interpolation and artificial intelligence (AI) methodologies for battery data expansion and validation [2][3]. However,

the complexity and time-consuming nature of traditional data collection (testing) methods can hinder the rapid advancement of EV technology. By incorporating GenAI methodologies, this study aims to extend traditional methods and streamline battery data augmentation and validation processes. This complementary approach accelerates EV system development process while maintaining the accuracy and reliability of traditional methods [4].

To improve the robustness of physics-based models, comprehensive data over different C-rates and temperatures are required. GenAI approach play a crucial role in deriving extrapolated properties, especially in scenarios where testing all possible C-rates and temperature profiles might be impractical or time-consuming.

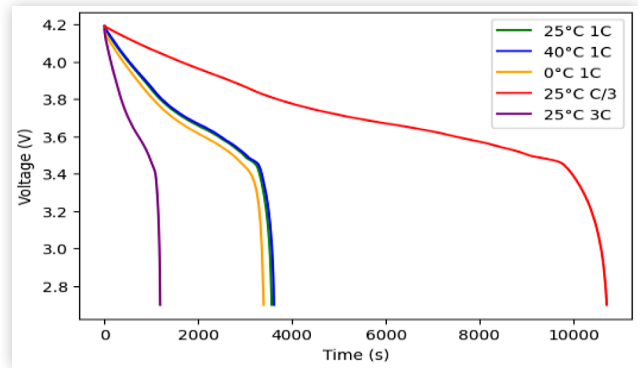
Proposed methodology entails a comparative analysis of three distinct GenAI models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Time-based Generative Pre-Trained Transformers (Time-GPT) for cyclic aging data. The dataset is derived from cyclic aging tests comprising one Full Cycle Equivalents (FCEs) with Constant Current (CC) charge/discharge currents at varying C-rates and temperatures. Additionally, the dataset includes cell data from different manufacturers, which are characterized by different specifications, chemical compositions and capacities.

This study evaluates the effectiveness of the GenAI engine in generating data with different C-rates from laboratory-measured data with few known C-rates [5][6]. Three different GenAI models are analyzed with reference to the metrics such as accuracy, consistency, and volume of data, contributing to the development of data-centric products tailored to battery health analysis. Through this collaborative synergy, the study catalyzes the evolution of EV technology and improves its efficiency, reliability, and sustainability.

Example Data

Cycle aging datasets represent a common data form in modern battery research and are used to empirically investigate the influence of various factors, such as charge current, discharge current, and temperature, on the battery's capacity retention over cycling. This dataset is from Li-Ion Cell (capacity 10Ah), encompasses measurements of current, voltage, and temperature recorded throughout the cycling process, alongside capacity and internal resistance or impedance metrics for each cycle. Batteries of varying capacities, specifications, and chemistries are subjected to cycling tests under distinct environmental temperatures (e.g. 0°C, 25°C, 45°C), using a standardized constant current charging and discharging (CC-charge, CC-discharge) procedure as shown in Figure 1 for cell C5C085. Furthermore, various discharge regimes are implemented to comprehensively assess the performance of the battery under different operating conditions.

FIGURE 1 Voltage characteristics of 10Ah cell.



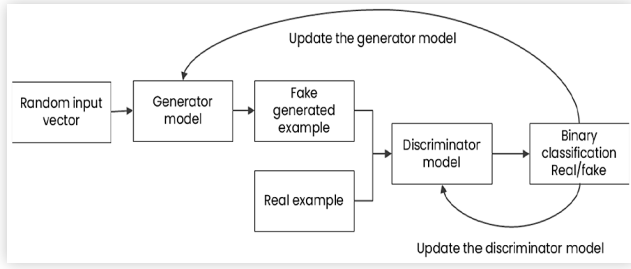
Background: Introduction to Generative AI Models

Generative Artificial Intelligence (GenAI) stands at the forefront of a transformative shift in data expansion, enabling unprecedented capabilities for synthesizing complex datasets from diverse domains. By leveraging advanced machine learning algorithms, GenAI techniques empower researchers to generate synthetic data that accurately reflect real-world phenomena. In the field of battery research and electric vehicle (EV) technology, GenAI is proving to be a promising ally to complement traditional data collection methods. Rather than replacing existing approaches, GenAI serves as a catalyst to overcome the challenges posed by the scarcity of comprehensive battery datasets. Through the integration of GenAI methodologies, researchers can augment traditional methods, propelling rapid advancements in battery modeling, simulation, and optimization [7]. This collaborative synergy between GenAI and traditional methods not only expands the scope of data generation but also improves the accuracy and reliability of predictive models, thereby accelerating the pace of innovation in EV technology.

Present scope explores three such GenAI models, (A) GAN (Generative Adversarial Networks) [8], (B) VAE (Variational Autoencoders) [9], and (C) Time-GPT (Time-based Generative Pre-Trained Transformers), which are explained further in following sections.

A. GAN

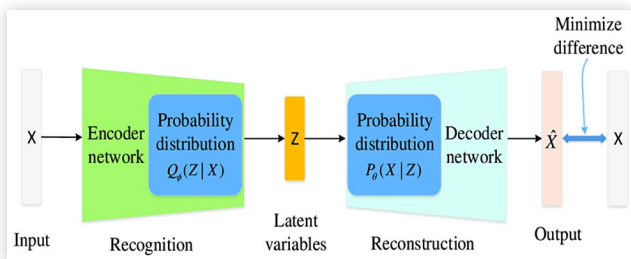
One of the pioneering approaches in GenAI is the Generative Adversarial Network (GAN) model. GANs consist of two neural networks, namely the generator and the discriminator, engaged in a competitive training process. The generator network learns to generate realistic data samples, while the discriminator network aims to distinguish between real and synthetic data. During training, GAN takes in two inputs, random noise data and an input data that is not labelled. Using these two inputs,

FIGURE 2 GAN architecture.

it generates data that resembles the input data as shown in Figure 2 [10]. Since all the inputs to the GAN are unlabeled, GAN is a type of unsupervised machine learning. Through iterative training, GANs achieve remarkable proficiency in generating high-fidelity data representations. In the realm of battery research, GANs exhibit potential in synthesizing diverse battery datasets encompassing various operating parameters such as charge-discharge cycles, temperature profiles, and chemical compositions. Moreover, GANs offer the flexibility to adapt and refine their data generation capabilities through continuous learning, making them indispensable tools for accelerating battery technology development.

B. VAE

Another prominent GenAI approach is the Variational Autoencoder (VAE) model, renowned for its ability to learn rich probabilistic representations of data. VAEs consist of an encoder network that maps input data to a latent space and a decoder network that reconstructs data from latent representations. By optimizing a variational lower bound, VAEs learn a latent space representation that captures the underlying structure of the data distribution as shown in Figure 3 [11]. In the domain of battery research, VAEs offer a principled framework for generating diverse battery datasets while preserving essential features and correlations. With their probabilistic nature, VAEs facilitate uncertainty quantification in data generation, providing valuable insights into the robustness and reliability of synthetic datasets. Furthermore, VAEs can seamlessly integrate with Bayesian inference techniques, enabling probabilistic reasoning and decision-making in battery system design and optimization.

FIGURE 3 VAE architecture.

C. Time-GPT

A cutting-edge addition to the GenAI arsenal is the Time-based Generative Pre-Trained Transformer (Time-GPT) model, inspired by the success of transformer architectures in natural language processing tasks. Time-GPT extends the transformer architecture to temporal data sequences, enabling effective modeling of time-dependent phenomena. By leveraging self-attention mechanisms, Time-GPT captures long-range dependencies and temporal correlations in sequential data. In the context of battery research, Time-GPT holds promise in generating time-series datasets that encapsulate the dynamic behavior of batteries over extended periods. With its ability to capture temporal patterns and trends, Time-GPT facilitates the generation of realistic battery data that reflects the complex interplay between charging cycles, aging effects, and environmental factors. Furthermore, Time-GPT's pre-trained representations can be fine-tuned on specific battery datasets, enhancing its adaptability and effectiveness in generating tailored synthetic data for diverse applications in EV technology and battery analytics.

Training Procedure & Comparative Analysis

The proprietary dataset used is of 80GB consisting of automotive grade Li-ion cells from different manufacturers spanning different cell chemistries (LFP & NMC) and cell specifications (capacity ranging between 2.5Ah to 55Ah). The data is compiled and preprocessed for training purposes according to different C-rates and temperatures. The preprocessing steps include handling missing values and checking data integrity. Each data set in plain text file format contains eight features inclusive of current, voltage, time, temperature, capacity and other derived parameters and spans $\sim 10^5$ records. To enhance the accuracy of battery data models, which are increasingly demanded by Original Equipment Manufacturers (OEMs), it is imperative to train them on a diverse range of C-rates. Traditionally, battery cell testing has been limited to a few fixed C-rates but leveraging GenAI models allow for the capture of data reflecting a wider array of charging rates, resembling real-world scenarios. In this study for NMC Li-ion cell, GenAI models are trained on a dataset comprising few fixed C-rates on a high performing computer with 6 GB GPU, and successfully generated data for intermediate and extrapolated C rates, capturing the voltage characteristic trends crucial for battery modeling. The training data used include 100% of the available C-rate data. However, for validation purposes, cross-validation technique is employed where data from one C-rate is omitted during training and used as the test dataset. This process is repeated for each C-rate, resulting in 90% of the data being used for training and 10% for testing in each iteration.

FIGURE 4 GAN model validation and prediction characteristics (a) Model validation (b) Validation error (c) Prediction for given C-rate

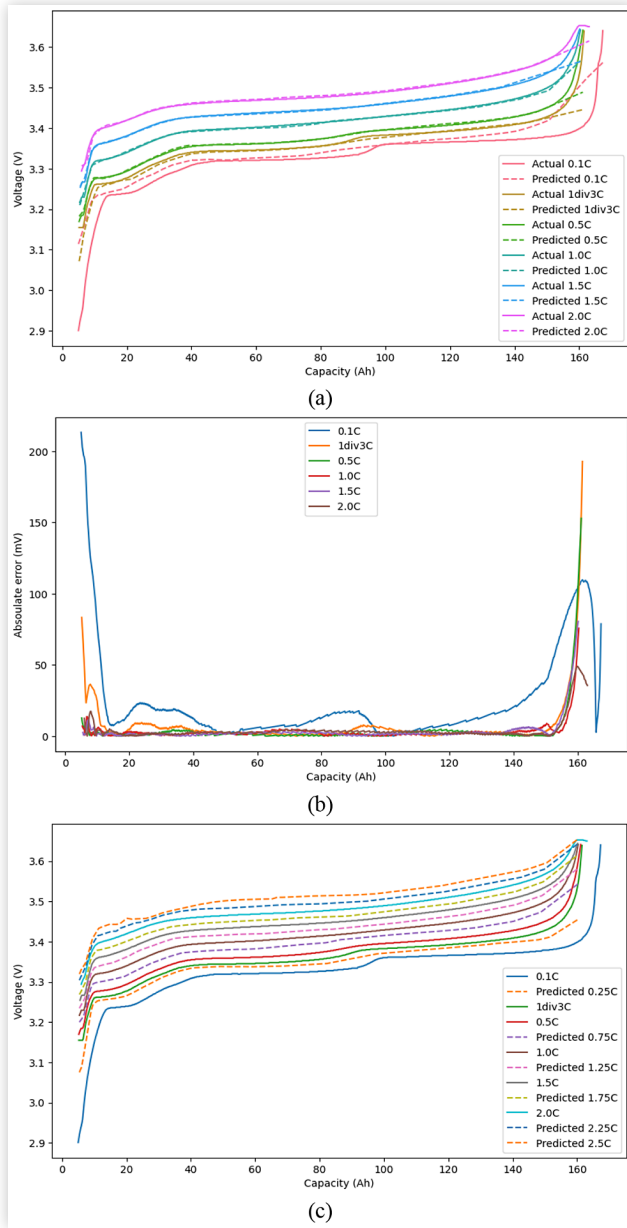


FIGURE 5 VAE model validation and prediction characteristics (a) Model validation (b) Validation error (c) Prediction for given C-rate

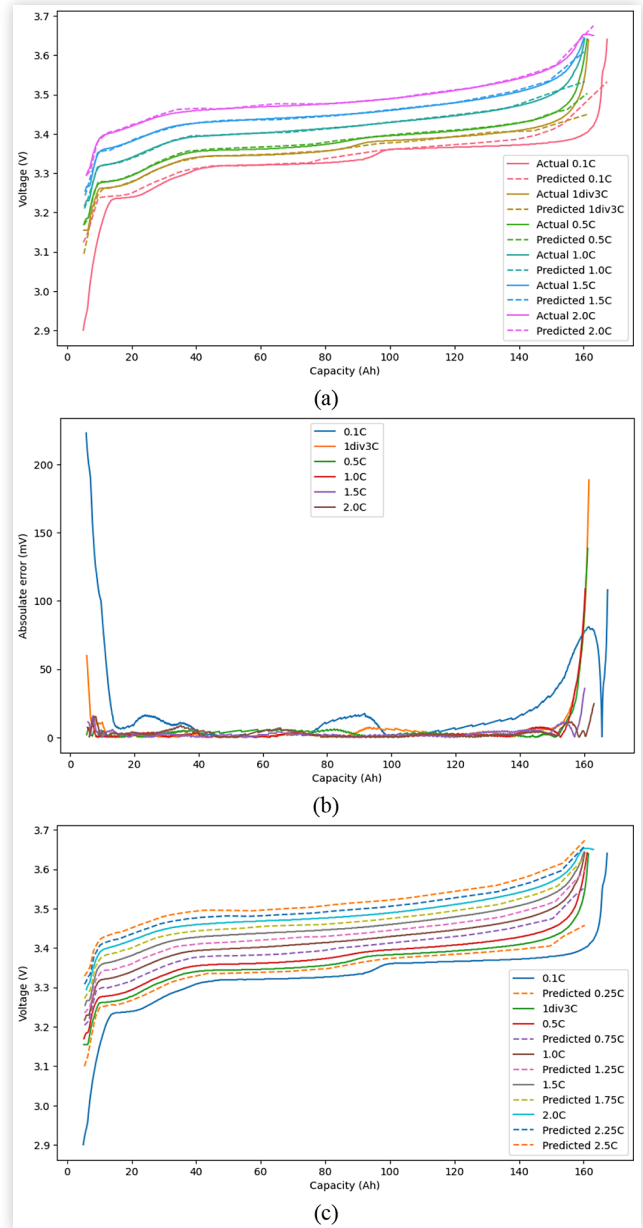


Figure 4, Figure 5, and Figure 6 illustrate the results obtained from the GAN, VAE, and Time-GPT models, respectively. These figures provide a detailed analysis of the models' capabilities in interpolating and extrapolating voltage behaviors across various charging rates, along with comprehensive model validation results and validation error graphs, highlighting the performance and accuracy of each model under different conditions. This highlights the potential of employing sophisticated generative models to enhance battery data models by incorporating insights from diverse charging scenarios. The models underwent training to predict voltage corresponding to distinct capacities and C rates. The Adam optimizer is used for every model, with a learning rate

adjustment strategy where the rate decreases if the loss does not decrease for a specified number of epochs. This technique, known as ReduceLROnPlateau, is implemented to improve model convergence. During training process, extending the K-fold technique, one of the six provided C rates and capacities is systematically omitted, and the models are trained on the remaining five sets of data. Subsequently, voltage predictions are generated for the omitted data, repeating this procedure iteratively for the entirety of the dataset. The training results are tabulated in Table 1 and training loss is shown in Figure 7. Notably, errors are minimized within the capacity range of 20% to 80%. Furthermore, the models demonstrated adeptness in capturing voltage characteristics across most C rates,

FIGURE 6 Time-GPT model validation and prediction characteristics (a) Model validation (b) Validation error (c) Prediction for given C-rate

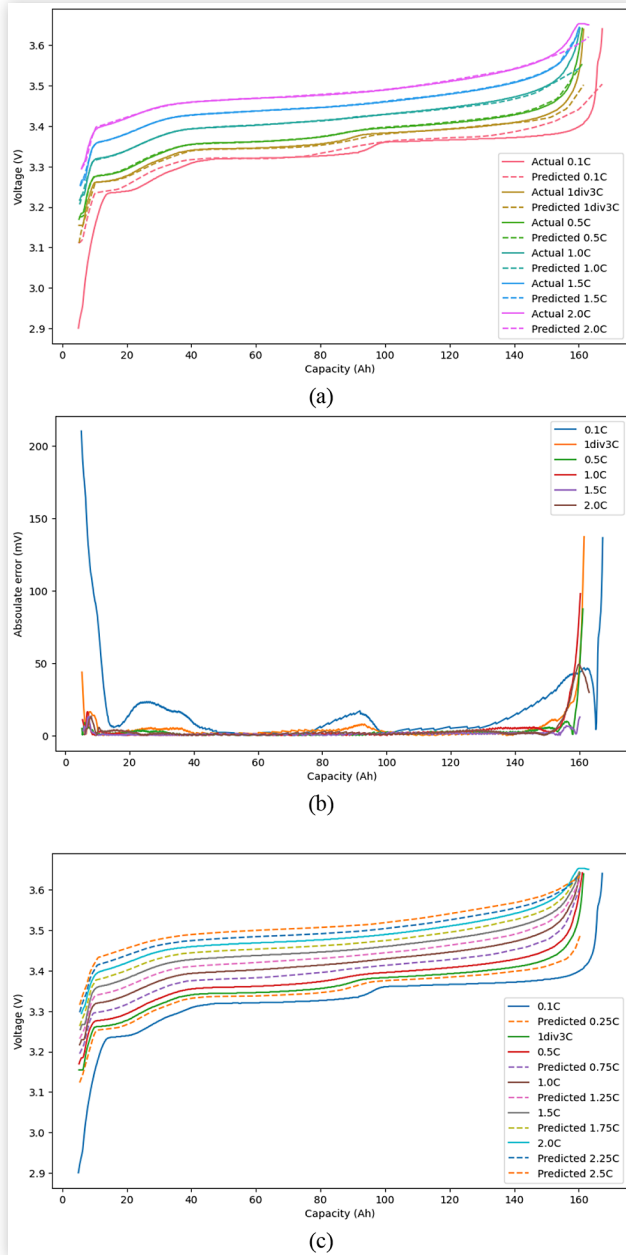
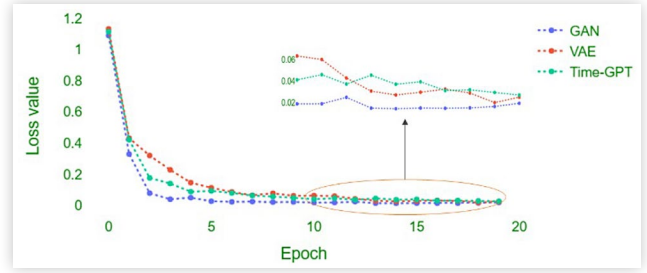


TABLE 1 Training performance of the models

Training parameters	GAN	VAE	Time-GPT
Execution time	1.5hour	1.2hour	0.8hour
Data size	5GB		
Learning rate	0.001		
Training Loss	Starting loss: 1.884 Ending loss: 0.0015	Starting loss: 1.1036 Ending loss: 0.0021	Starting loss: 1.09 Ending loss: 0.0009

FIGURE 7 Training loss



albeit encountering challenges with the 0.1C rate. Among the experiments conducted, the Time-GPT model exhibited optimal performance for the 1.5C rate. Subsequently, the models underwent comprehensive training using all available input data, enabling the extension of voltage predictions for intermediate and extrapolated C rates. Notably, the voltage characteristics observed in the extended results closely aligned with both intermediate and extrapolated trends.

Following an exhaustive comparative analysis of the three models across various parameters, it is observed that the Time-GPT model perform quite better than other two models. Specifically, it demonstrate a superior ability to capture the non-linear voltage trends with greater accuracy. This is quite expected as Time-GPT model typically demonstrate strong performance where time series data is employed for extension or validation reasons [13].

Conclusion

The comparative analysis of GenAI models to extend battery data for modelling shows the capability of using GenAI technology in an effective and innovative way. Different GenAI models are trained and validated on battery cell testing data to expand the data: GAN, VAE, and Time GPT. The data consists of various battery parameters out of which this paper focuses on voltage characteristics.

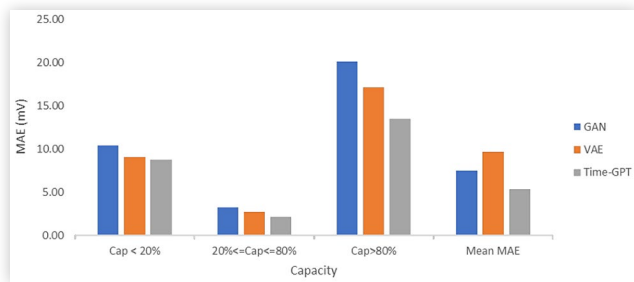
Table 2 delineates the validation error characteristics of the models, where Mean Absolute Error (MAE) is computed across three distinct capacity intervals based on typical Li-ion cell non-linearity characteristics: <20%, 20% to 80%, and >80%. Additionally, the mean MAE across these intervals is determined. Consolidation of these error characteristics over constant C-rate is presented in Figure 8.

Upon scrutinizing the results alongside other operating parameters of the three models, it became evident that Time-GPT exhibited superior performance compared to the other models and observation is similar for variety of Li-ion cell chemistry and cell specifications.

The application of generative AI for battery data extension/expansion is relatively a new concept, and therefore quite challenging to explore established benchmarks for better comparison of the methodology proposed. However, some of the parallel works [12] exhibit

TABLE 2 Mean absolute error MAE (mV) characteristics

C - Rate	GAN model				VAE model				Time-GPT model			
	Cap < 20%	20% < = Cap < = 80%	Cap > 80%	Mean MAE	Cap < 20%	20% < = Cap < = 80%	Cap > 80%	Mean MAE	Cap < 20%	20% < = Cap < = 80%	Cap > 80%	Mean MAE
0.1	42.3	8.64	53.8	23.4	39.3	5.71	41.3	28.8	40	5.42	31.4	16.7
0.33	11.2	2.63	19.3	6.94	3.98	2.34	20.6	8.96	4.91	2.26	13.9	4.66
0.5	2.39	1.65	14.9	3.93	2.13	3.26	14.6	6.65	2.4	1.18	8.29	2.56
1	2.21	1.91	8.96	3.08	2.1	1.25	14.2	5.83	1.58	1.34	129	3.2
1.5	1.66	1.62	12.5	3.34	2.88	1.67	6.85	3.8	1.08	0.95	2.46	1.21
2	2.6	2.71	11.3	4.18	4.09	2.1	5	3.73	2.65	1.41	11.9	3.44
Mean MAE	10.4	3.19	20.1	7.48	9.07	2.72	17.1	9.62	8.77	2.09	13.5	5.29

FIGURE 8 Mean MAE of all C rates for 3 models.

about similar error characteristics showing permissible mean absolute error (MAE) as 1.5%. GenAI based approach illustrate the error margins quite well withing this expected permissible level.

The work presented is illustrated with NMC Li-ion cell, and similar characteristics of the approach are observed for LFP Li-ion cell and specifications proving the applicability of approach independent of cell chemistry and specifications.

References

- Korosec, K., "How Generative AI Is Creeping into EV Battery Development," Retrieved from Techcrunch, 2023, <https://techcrunch.com/2023/10/14/how-generative-ai-is-creeping-into-ev-battery-development>.
- Jianga, L., Hua, C., Jia, S., Zhaoa, H. et al., "Generating Comprehensive Lithium Battery Charging Data with Generative AI," 2024, <https://arxiv.org/pdf/2404.07577>.
- Pyne, M., Yurkovich, B.J., and Yurkovich, S., "Generation of Synthetic Battery Data with Capacity Variation," in *IEEE Conference on Control Technology and Applications (CCTA)*, 2019, IEEE.
- Singh, J.P., "Ai-Driven Solutions For Next- Generation Electric Vehicles," ResearchGate. (PDF) Ai-Driven Solutions for Next- Generation Electric Vehicles, www.researchgate.net.
- Topan, P.A. et al., "State of Charge (SOC) and State of Health (SOH) Estimation on Lithium Polymer Battery via Kalman Filter," in *2016 2nd International Conference on Science and Technology-Computer (ICST)*, 2016, IEEE.
- Ramzan, F., Sartori, C., Sartori, C., Consoli, S. et al., "Generative Adversarial Networks for Synthetic Data Generation in Finance: Evaluating Statistical Similarities and Quality Assessment," *AI | Free Full-Text | Generative Adversarial Networks for Synthetic Data Generation in Finance: Evaluating Statistical Similarities and Quality Assessment*, 2023, www.mdpi.com.
- Sohn, K., Lee, H., and Yan, X., "Learning Structured Output Representation using Deep Conditional Generative Models," in *Advances in Neural Information Processing Systems 28 (NIPS 2015)*, 2015.
- Packt_Pub, "Inside the Generative Adversarial Networks (GAN) Architecture," Medium, 2019, https://medium.com/@Packt_Pub/inside-the-generative-adversarial-networks-gan-architecture-2435afbd6b3b.
- Rocca, J., "Understanding Variational Autoencoders (VAEs)," 2019, <https://towardsdatascience.com/understanding-variational-autoencoders-vaes-f70510919f73>.
- Bhavani, V., "Comprehensive Overview of GAN Technology: Architecture, Training, and Applications," *Comprehensive Overview of GAN Technology: Architecture, Training, and Applications*, 2024, www.ijraset.com.

11. Wei, R., Garcia, C., El-Sayed, A., Peterson, V. et al., "Variations in Variational Autoencoders - A Comparative Evaluation," 2020.
12. Jonghyeon, K. and Kowal, J., "Development of a Matlab/Simulink Model for Monitoring Cell State-of-Health and State-of-Charge via Impedance of Lithium-Ion Battery Cells," 2022, www.mdpi.com
13. Garza, A., Challu, C., and Mergenthaler-Canseco, M., "TimeGPT-1," 2024, <https://arxiv.org/pdf/2310.03589>.

Contact Information

Mailing address: FEV smart mobility center (Engineering Analytics Department), Survey no.2, Hissa No 7/1, Baner, Pune, Maharashtra 411045.

hivarkar@fev.com

Telephone number: +918888795482